📘 1️⃣ Servlet Life Cycle Methods

🔹 init()

What it does:

- Called only once when servlet is loaded.
- Used for initialization work.
- Reads init parameters.
- Establish DB connection.

When it runs:

Client Request → Servlet Loaded → init() → service()

Example:

```
public void init() {
   System.out.println("Servlet Initialized");
}
```

Exam Point:

Used to allocate resources.

🔹 service()

What it does:

- Handles every request.
- Internally calls:
    - doGet()
    - doPost()

You normally don't override service().

🔹 destroy()

What it does:

- Called once before servlet is removed.
- Used to:
    - Close DB connection
    - Release resources

📘 2️⃣ HTTP Methods

🔹 doGet(HttpServletRequest req, HttpServletResponse res)

What it does:

- Handles GET request.
- Reads data from URL query string.
- Used for:
  - Fetching data
  - Search
  - Simple navigation

Example:

```
String name = request.getParameter("name");
```

🔹 doPost(HttpServletRequest req, HttpServletResponse res)

What it does:

- Handles POST request.
- Reads form data from request body.
- Used for:
  - Login
  - Registration
  - Form submission

📘 3️⃣ HttpServletRequest Methods

🔹 getParameter(String name)

What it does:

- Returns form value as String.
- Used to read input fields.

Example:

```
String username = request.getParameter("username");
```

🔹 getParameterValues(String name)

Returns multiple values (checkbox).

- **getHeader(String name)**

Returns specific HTTP header.

Example:

```
request.getHeader("User-Agent");
```

- **getMethod()**

Returns request method:

GET or POST

- **setAttribute(String name, Object value)**

Stores data in request object.

Used for:

- **Passing data to JSP**
- **Used with forward()**

- **getAttribute(String name)**

Retrieves stored attribute.

- **getRequestDispatcher(String path)**

Returns RequestDispatcher object.

Used to forward request.

📘 4️⃣ HttpServletResponse Methods

- **getWriter()**

Returns PrintWriter.

Used to send response to browser.

Example:

```
PrintWriter out = response.getWriter();

out.println("Hello");
```

- **setContentType(String type)**

Sets MIME type.

Example:

response.setContentType("text/html");

🔹 setStatus(int code)

Sets HTTP status code.

Example:

response.setStatus(200);

🔹 sendRedirect(String location)

What it does:

- Sends response to browser.
- Browser makes new request.
- URL changes.
- Data lost.

Example:

response.sendRedirect("home.jsp");

📘 5️⃣ RequestDispatcher Methods

🔹 forward(request, response)

What it does:

- Transfers request to another resource.
- Same request object.
- Data preserved.
- URL does NOT change.

Example:

RequestDispatcher rd = request.getRequestDispatcher("result.jsp");

rd.forward(request, response);

🔹 include(request, response)

Includes output of another resource inside current response.

Used for:

- Header.jsp
- Footer.jsp

📘 6️⃣ HttpSession Methods

🔹 **getSession()**

Creates or returns existing session.

🔹 **setAttribute(String name, Object value)**

Stores user data in session.

Example:

```
session.setAttribute("user", "Harshal");
```

🔹 **getAttribute(String name)**

Retrieves session data.

🔹 **invalidate()**

Destroys session.

Used for logout.

📘 7️⃣ ServletConfig Methods

🔹 **getInitParameter(String name)**

Returns servlet-specific parameter from web.xml.

🔹 **getServletName()**

Returns servlet name.

📘 8️⃣ ServletContext Methods

🔹 **getInitParameter(String name)**

Returns application-wide parameter.

🔹 **setAttribute(String name, Object value)**

Stores data globally.

🔹 getAttribute(String name)

Retrieves global data.

📘 9️⃣ @WebServlet Annotation

What it does:

- Replaces web.xml mapping.

Example:

```
@WebServlet("/add")
public class AddServlet extends HttpServlet {
}
```

🎯 Quick Comparison Table

Feature	forward()	sendRedirect()
Side	Server	Client
URL Change	No	Yes
Data Preserved	Yes	No
Request Type	Same	New